

RayGlides 3W/4W EV Kit Cooling Controller - Component Selection Rationale (Optimized for ESP32-S3-N16R8)

This document provides a detailed engineering review of the primary component selections, package decisions, and alternative source allocations.

1. Primary Component Choices

A. Power Regulator: LM5017MR/NOPB (Texas Instruments)

- * **Alternative Part**: LMR16006YQ5DDCTQ1 (Automotive-grade, 60V, SOT-23-6)
- * **Package**: SO-PowerPAD-8 (with exposed bottom thermal pad)
- * **Voltage/Current Rating**: 100V peak input voltage, 600mA maximum output current
- * **Key Specifications**: Constant On-Time (COT) control loop, integrated high-side and low-side N-Channel switches, UVLO threshold.
- * **Selection Rationale**: In 3W/4W EV traction power networks, transient voltage spikes from the motor controller switching and regenerative braking cycles can exceed 80V . The LM5017's 100V operating limit provides a massive safety margin, preventing regulator breakdown without requiring large shunt clamp resistors.

B. Switching MOSFET: AO3400A (Alpha & Omega Semiconductor)

- * **Alternative Part**: BSS138 (50V, SOT-23, lower current) or PMV16XNR (30V, SOT-23, automotive)
- * **Package**: SOT-23
- * **Voltage/Current Rating**: 30V maximum drain-source voltage, 5.7A maximum continuous drain current
- * **Key Specifications**: $R_{DS(on)} < 33\text{m}\Omega$ at $V_{GS} = 3.3\text{V}$, gate threshold voltage $V_{GS(th)} \approx 1.0\text{V}$.
- * **Selection Rationale**: The ESP32-S3 microcontroller outputs a 3.3V logic signal. Standard MOSFETs require 5V or 10V gate drive to saturate. The AO3400A is a true logic-level MOSFET that saturates at 2.5V gate voltage. The low $R_{DS(on)}$ at 3.3V minimizes conduction heating, allowing the SOT-23 package to switch the 150mA fan safely.

C. Temperature Sensor: LM35DT/NOPB (Texas Instruments)

- * **Alternative Part**: MCP9700A-E/TO (Active analog, 10mV/deg C , 500mV offset at 0 deg C)
- * **Package**: TO-220-3 (Metal tab package)
- * **Accuracy**: $\pm 0.5^\circ\text{C}$ at room temperature, calibrated directly in Celsius
- * **Key Specifications**: 10 mV/deg C scale factor, works from 4V to 30V VCC .
- * **Selection Rationale**: The TO-220 package can be directly bolted to a heatsink using an M3 screw. LM35 has a linear output ($10\text{mV/}^\circ\text{C}$), which requires no complex polynomial scaling in the firmware, unlike thermistors.

D. CAN Transceiver: TJA1050T/3 (NXP Semiconductors)

- * **Alternative Part**: SN65HVD230 (3.3V powered CAN transceiver) or MCP2551

- * ****Package****: SOIC-8
- * ****Voltage Rating****: 5V VCC power supply, 1 Mbps high-speed transmission
- * ****Selection Rationale****: The ESP32-S3 has a built-in TWAI (CAN) controller. The TJA1050T interfaces directly with ESP32-S3 GPIO 9 and 10 to establish robust differential CANH/CANL lines. Powered from the stable 5V rail, it provides high noise immunity and electrostatic discharge (ESD) protection for automotive CAN bus environments.

E. ADC Voltage Reference: LM4040AIM3-1.0/NOPB (Texas Instruments)

- * ****Alternative Part****: ADR510ARTZ (1.0V precision shunt voltage reference)
- * ****Package****: SOT-23
- * ****Key Specifications****: 1.0 V output, 0.1% accuracy, low temperature coefficient.
- * ****Selection Rationale****: The ESP32-S3's internal ADC reference is unstable and exhibits non-linear behavior under RF loading. Using the LM4040AIM3-1.0 reference diode enables precise calibration of the ADC, mapping the LM35's 0-1.0V (0 deg C to 100 deg C) linear analog voltage with maximum resolution.

F. ESD Protection: USBLC6-2SC6 (STMicroelectronics)

- * ****Package****: SOT-23-6
- * ****Selection Rationale****: A low-capacitance rail-to-rail ESD protection array placed directly on the USB-C D+/D- lines prevents electro-static discharges from the programming cable from damaging the ESP32-S3 native USB peripheral.